PDC Reconstruction Literature Survey and SAMURAI Terminal Air-MS Gap Analysis

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Contents

1 TL;DR

This document surveys the published reference frame for the SAMURAI PDC1/PDC2 drift-chamber chain and contrasts it with our current Geant4 reconstruction performance, in order to localise the air multiple-scattering gap that drives our PDC2 spatial resolution one order of magnitude above design.

- **Reference frame.** The PDC1/PDC2 chain has no dedicated peer-reviewed publication. The canonical references are the SAMURAI overview paper by Kobayashi et al., Nucl. Instr. Meth. B **317**, 294–304 (2013) §3.4[1], the RIKEN internal technical note `Detector-PDC.pdf`[2], and the Kahlbow PhD thesis (TU Darmstadt, defended 2019; TUprints pub. 2021)[3] which documents an independent reconstruction of the same hardware.
- **Design specifications.** Combining Kobayashi 2013 §3.4[1] with the SAMURAI 2012 construction proposal[4], the PDC1/PDC2 chain targets 1 mm position resolution, 2 mrad angular resolution, and 0.8% relative momentum resolution ($\Delta p/p$). Commissioning RI-beam runs reached $\Delta p/p \approx 1/1500$[1]; the original proton-mode design value is $\Delta p/p \approx 1/700$ as quoted in the RIBF-TAC05 (2005) construction proposal[5].
- **Our measured σ_y(PDC2).** In our Geant4 + reconstruction chain we observe $\sigma_y(\text{PDC2}) =$ 12.83–14.93 mm in the all-air configuration, 6.80–8.21 mm when Region A (dipole cavity + beam pipe) is set to vacuum (`beamline_vacuum`), and approximately 1.0 mm in the all-vacuum configuration[6]. The realistic-air values sit roughly an order of magnitude away from the 1 mm hardware design specification[1].
- **Air MS dominates and the configuration choice matters.** A quadrature decomposition[6] attributes the all-air budget to $\sigma_A \approx 11.19$ mm from Region A (cavity + beam pipe, ∼4.3 m) and $\sigma_B \approx 7.43$ mm from Region B (Exit Window → PDCs, ∼2 m). The stage D inputs from the same memo[7] give $\sigma_y(\text{PDC2}) = 7.49$ mm if Region A is vacuum and $\sigma_y(\text{PDC2}) = 13.50$ mm if both Region A and Region B are air. Whether Region A is air or vacuum during the experiment is therefore a configuration question that selects between these two values; this document does not assert which one corresponds to the actual run.

See §6 for proposed action items.

2 The problem: air multiple scattering at the SAMURAI terminal

2.1 Geometry of the proton flight path

The deuteron target sits at world coordinates $(-12.488, 0.0009, -1069.458)$ mm, which corresponds to yoke-local coordinates $(-545.5, 0, -919.9)$ mm under the -30° yoke rotation about Y[8]. Because the SAMURAI dipole cavity (`vchamber_box`) extends over $z \in [-1750, +1750]$ mm

[1] T. Kobayashi et al., Nucl. Instr. Meth. B **317**, 294–304 (2013), DOI: 10.1016/j.nimb.2013.05.089.

[2] RIKEN Nishina Center, *Detector-PDC* technical note, `https://www.nishina.riken.jp/ribf/SAMURAI/image/Detector-PDC.pdf`.

[3] J. Kahlbow, PhD thesis, TU Darmstadt (defended 2019; TUprints publication year 2021), TUprints 9192, `https://tuprints.ulb.tu-darmstadt.de/9192/`.

[4] SAMURAI Collaboration, *SAMURAI Construction Proposal* (2012), `https://ribf.riken.jp/SAMURAI/120425_SAMURAIConstProp.pdf`.

[5] SAMURAI Collaboration, *RIBF-TAC05 SAMURAI construction proposal* (2005), `https://ribf.riken.jp/RIBF-TAC05/10_SAMURAI.pdf`.

[6] See `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §4.

[7] See `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §6.

[8] See `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §1.1–§1.3 for coordinates, yoke rotation, and Region A/B definitions.

in yoke-local frame, the target at $z = -919.9$ mm is *physically inside the dipole cavity* rather than upstream of it. Two regions therefore separate the target from the PDCs. **Region A** (≈ 4.3 m) covers the in-cavity flight from the target through the cavity exit (2.67 m) and onwards through the downstream beam-pipe (`VacuumDownstream`, ≈ 1.6 m) up to the Exit Window; this region is controlled by the Geant4 macro switch `/samurai/geometry/BeamLineVacuum`. **Region B** (≈ 2.0 m) covers free flight in the world volume from the Exit Window at world $(170.5, 0, 3187.5)$ mm to PDC1 at $(700, 0, 4000)$ mm (0.97 m) and on to PDC2 (1.00 m), gated by `/samurai/geometry/FillAir`. Inside Region A the proton bends $\sim 30°$ in the 1.15 T field, but the arc-length vs. straight-line difference stays below 10%[8].

2.2 Measured σ values

Table 1 reproduces the robust half-width $\sigma = (p_{84} - p_{16})/2$ measured at PDC1 and PDC2 across three vacuum/air configurations and three truth momenta, with $|\vec{p}| = 627$ MeV/c protons.

Condition	(p_x, p_y) [MeV/c]	N	σ_x(P1)	σ_y(P1)	σ_x(P2)	σ_y(P2)	σ_{θ_x}	σ_{θ_y}
all_air	$(0, 0)$	50	3.071	9.736	3.669	12.826	12.279	9.240
all_air	$(100, 0)$	50	3.838	10.492	5.377	12.751	15.768	8.277
all_air	$(0, 20)$	50	3.372	13.039	4.065	14.927	21.709	7.846
beamline_vacuum	$(0, 0)$	50	1.528	4.061	2.086	6.801	11.834	5.910
beamline_vacuum	$(100, 0)$	50	2.067	4.888	3.060	8.211	8.100	5.786
beamline_vacuum	$(0, 20)$	49	1.667	4.789	2.363	7.459	14.289	5.421
all_vacuum	$(0, 0)$	50	0.147	0.515	0.447	0.947	6.798	2.558
all_vacuum	$(100, 0)$	50	0.154	0.364	0.416	0.999	4.075	1.956
all_vacuum	$(0, 20)$	50	0.174	0.411	0.396	1.002	6.913	2.299

Table 1: σ values from Geant4 + reconstruction ensemble; σ in mm, σ_θ in mrad. Robust half-width $(p_{84} - p_{16})/2$. Source: `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §4.

2.3 Gap to design specifications

The realistic-air spatial σ_y(PDC2) sits roughly an order of magnitude away from the 1 mm hardware design figure, and only the all-vacuum ablation reaches the design floor. Table 2 summarises the comparison; angular and momentum entries follow the same structure.

Quantity	Design / commissioning	Our current state	Source
Position σ at PDC	1 mm	~ 13 mm (all-air) / ~ 7.5 mm (A=vac) / ~ 1.0 mm (all-vac)	Kobayashi 2013 §3.4[1] + memo §4[6]
Angular σ	2 mrad	5.4–9.2 mrad (σ_{θ_y}, all-air & `beamline_vacuum`) / 2.0–2.6 mrad (all-vacuum)	Kobayashi 2013 §3.4[1] + memo §4[6]
RI commissioning $\Delta p/p$	1/1500	not measured in this ablation	Kobayashi 2013[1]
Proton-mode design $\Delta p/p$	$\sim 1/700$	not measured in this ablation	RIBF-TAC05 (2005)[5]

Table 2: PDC reconstruction performance: design / commissioning targets vs. our current Geant4 + reconstruction state. Position and angular rows compare against the Kobayashi 2013 §3.4 hardware specification; the two $\Delta p/p$ rows are not yet propagated end-to-end in our ablation, so we have σ_y(PDC2) but no single $\Delta p/p$ figure to report.

2.4 Why this matters for the science floor

The PDC2 σ_y entries above are not the end of the story: they propagate backwards through the Runge–Kutta tracker and the SAMURAI magnetic field map to a target-momentum uncertainty $\sigma_{p,\text{target}}$, which is the reconstruction science floor for any experiment in which the target sits inside the magnet and the outgoing proton is measured downstream (target → magnet → PDC). This document does not re-derive $\sigma_{p,\text{target}}$ propagation values; the per-method numerical breakdown lives in the RK error-analysis deep-dive memo[9].

3 PDC reconstruction reference frame

3.1 Hardware foundations

The PDC1/PDC2 hardware specification is documented across four canonical sources: Kobayashi 2013 §3.4[1], the RIKEN internal `Detector-PDC.pdf` tech note[2], the SAMURAI 2012 construction proposal[4], and the older RIBF-TAC05 (2005) construction proposal[5]. The proton-arm magnet is the SAMURAI superconducting H-shape dipole described by Sato et al., providing a 7 Tm bending power, a maximum field of 3 T, an 80 cm pole gap and a 2 m pole face, with the field map computed in OPERA-3d / TOSCA[10]. The drift cells follow the Walenta geometry with an 8 mm drift distance, arranged as five planes (U/V/X/U/V) at $\pm 45°/0°$ over a 1700×800 mm^2 active area, filled with Ar + 25% iC_4H_{10} (or alternatively Ar + 50% C_2H_6) and read out via ∼810 channels[2]. The chambers sit at $\sim 59°$ on the proton arm with the design targets of 1 mm position resolution, 2 mrad angular resolution, and 0.8% relative momentum resolution, with commissioning RI runs reaching $\Delta p/p \approx 1/1500$[1] and a proton-mode design specification of $\Delta p/p \approx 1/700$[5]. The signal chain feeds an ASD shaper into AMT-VME TDCs at 0.78 ns/ch[5], where the AMT chip is the TDC ASIC originally described by Arai[11]. The boundary between the dipole-cavity vacuum (Region A) and the downstream air volume (Region B) is the SAMURAI Exit Window, which on the actual SAMURAI spectrometer is a metallic structure; our simulation models it as an iron (G4_Fe) flange with a Galactic-vacuum or air-filled central aperture (`libs/smg4lib/src/devices/ExitWindowC1Construction.cc`, `ExitWindowC2Construction.cc`). The dipole cavity itself is evacuated to a few Pa during operation, as documented for the original 2012 SAMURAI vacuum-system commissioning by Shimizu et al.[12]; we cite this reference only for the cavity-evacuated status, not for the exit-window material. The matching offline reconstruction pipeline is the RIKEN ROOT-based drift-chamber analysis described by Otsu et al.[13]. Together these four canonical sources cover the hardware geometry and design specifications, but none of them publishes a head-to-head proton-arm reconstruction performance paper; that gap is the motivation for §3.2 (algorithm spine) and §3.3 (performance achieved).

[9] See `docs/reports/reconstruction/momentum_reco/rk/rk_error_analysis_deep_dive_20260426_zh.pdf` for $\sigma_{p,\text{target}}$ propagation under MC, Profile, Laplace, MCMC, and MC-PDC.

[10] H. Sato et al., IEEE Trans. Appl. Supercond. **23**, 4500308 (2013), DOI: 10.1109/TASC.2012.2237225.

[11] Y. Arai, Nucl. Instr. Meth. A **453**, 365 (2000), DOI: 10.1016/S0168-9002(00)00658-6.

[12] Y. Shimizu, H. Otsu, T. Kobayashi, T. Kubo, T. Motobayashi, H. Sato, K. Yoneda, "Vacuum system for the SAMURAI spectrometer," Nucl. Instr. Meth. B **317**, 739–742 (2013), DOI: 10.1016/j.nimb.2013.08.051.

[13] H. Otsu et al., Nucl. Instr. Meth. B **376**, 175 (2016), DOI: 10.1016/j.nimb.2016.02.056.

Item	Value	Source
PDC location	$\sim 59°$ on proton arm	Kobayashi 2013 §3.4
Active area	1700×800 mm^2	Detector-PDC tech note
Drift cell	Walenta, 8 mm drift	Detector-PDC tech note
Plane configuration	U/V/X/U/V at $\pm 45°/0°$	Detector-PDC tech note
Primary gas	Ar + 25% iC_4H_{10}	Detector-PDC tech note
Alternative gas	Ar + 50% C_2H_6	Detector-PDC tech note
Channels	~ 810	Detector-PDC tech note
Readout TDC	AMT-VME, 0.78 ns/ch	RIBF-TAC05 2005 + Arai 2000
Magnet type	superconducting H-dipole	Sato 2013
Bending power	7 Tm	Sato 2013
Max field	3 T	Sato 2013
Gap	80 cm	Sato 2013
Pole face	2 m	Sato 2013
Field map source	OPERA-3d / TOSCA	Sato 2013
Design position σ	1 mm	Kobayashi 2013 §3.4
Design angle σ	2 mrad	Kobayashi 2013 §3.4
Design $\Delta p/p$	0.8%	Kobayashi 2013 §3.4
Commissioning RI $\Delta p/p$	1/1500	Kobayashi 2013
Proton mode $\Delta p/p$ design	$\sim 1/700$	RIBF-TAC05 2005

Table 3: PDC1/PDC2 + magnet + ExitWindow hardware specifications. Sources cited inline; full bibliographic entries in §7.

3.2 Algorithm spine

The published reference reconstruction chain for the SAMURAI proton arm follows a four-stage spine. **(i) Hits to track segments.** Drift times measured by the AMT-VME TDCs are converted to drift distances and combined into a straight-line track in each PDC plane; this drift-time-to-distance and per-plane line-fit pipeline is the RIKEN ROOT-based drift-chamber analysis described by Otsu et al.[13], deployed within the SAMURAIANA / `anaroot` framework (cite by repo path `anaroot/`). **(ii) Back-tracking through the magnet.** The PDC-plane track is propagated backwards through the SAMURAI superconducting H-dipole using a Runge–Kutta integrator over the OPERA-3d / TOSCA field map published by Sato et al.[10]; the field map itself is peer-reviewed, but the back-tracking algorithmic recipe is documented only in PhD theses (see point iii) and not in any peer-reviewed paper. **(iii) Optional Kalman smoothing.** A Kalman-filter smoother on the PDC tracks is the standard refinement, but *PDC-side Kalman smoothing has no peer-reviewed publication; the full methodology lives only in PhD theses.* The closest published paper, Santamaria et al., Nucl. Instr. Meth. A **905**, 138–148 (2018)[14], applies a Kalman filter to the MINOS TPC, not the PDCs. The PDC-side methodology is recorded across four PhD theses: Santamaria 2015 (Univ. Paris-Saclay)[15], Kahlbow 2021 (TU Darmstadt)[3], Lehr ca. 2023 (TU Darmstadt) `[unverified --- TUprints/HAL record not located]`[16], and Storck-Dutine 2023 (TU Darmstadt)[17]. **(iv) Output.** The propagated state at the target plane yields the target-frame momentum vector $\vec{p}_{\text{target}}$ used downstream. As context, several MINOS+SAMURAI groups generate their training and validation data using the NPTool simulation framework of Matta et al., J. Phys. G **43**, 045113 (2016)[18], but NPTool is a simulation

[14] C. Santamaria et al., Nucl. Instr. Meth. A **905**, 138–148 (2018), DOI: 10.1016/j.nima.2018.07.053.

[15] C. Santamaria, PhD thesis, Univ. Paris-Saclay (2015), HAL tel-01231191, `https://hal.science/tel-01231191`.

[16] C. Lehr, PhD thesis, TU Darmstadt (2023). `[unverified --- TUprints/HAL record not located in 2026-04-27 audit; placeholder pending direct confirmation.]`

[17] S. Storck-Dutine, PhD thesis, TU Darmstadt (2023), TUprints 23768, `https://tuprints.ulb.tu-darmstadt.de/23768/`.

[18] A. Matta et al., J. Phys. G **43**, 045113 (2016), DOI: 10.1088/0954-3899/43/4/045113.

framework, not the reconstruction path itself.

Our internal pipeline lives in `libs/analysis_pdc_reco/` (see also `smsimulator5.5/CLAUDE.md` for the repo-level guide) and follows the same Runge–Kutta back-tracking spine, with multiple selectable RK fit modes; one concrete example is `--rk-fit-mode fixed-target-pdc-only` as exercised in the MS ablation memo[6] §3. An alternative neural-network backend is provided under `scripts/reconstruction/nn_target_momentum/`, and an RK geometry filter (`libs/qmd_geo_filter/`, with the matching skill `skills/smsim-rk-geometry-filter/`) prunes obviously off-acceptance trajectories before the fit. We do not re-derive the per-method numerical performance here: the full status, including MC, Profile, Laplace, MCMC, and MC-PDC variants, is reported in the RK status memo at `docs/reports/reconstruction/momentum_reco/rk/rk_reconstruction_status_20260416_` and the deep-dive[9]. A detailed algorithm-by-algorithm comparison between the literature spine and our internal pipeline is deferred to §4.2 (Methodology gap).

3.3 Performance achieved across SAMURAI campaigns

Table 4 surveys the public SAMURAI papers that have used the PDC1/PDC2 chain and asks a single question: did the publication report a PDC-side σ_p/p figure that we can directly compare against our reconstruction? The answer for almost every entry is no — public papers report physics observables (cross sections, decay energies, momentum distributions) rather than the upstream tracker resolution. Cells are therefore explicitly marked "not reported" or "needs verification"; we deliberately do not invent numbers.

Experiment	Year	Channel	σ_p/p reported	Source flag	Citation
Commissioning RI	2012	RI ions	$\sim 1/1500$	reported	Kobayashi 2013[1]
SEASTAR-III ^{28}O	2017	(p,2p)/(p,pn)	not reported (0.2 MeV FWHM decay-E $\leftrightarrow$ inferred $\sim 1\%$)	inferred	Kondo Nature 620 (2023)[19]
^{52}Ca(p,pn)^{51}Ca	2017	(p,pn)	not reported	needs verification	Enciu PRL 129 (2022)[20]
^{30}F study	2017	(p,2p)	not reported	needs verification	Kahlbow PRL 133 (2024)[21]
^{50}Ar(p,pn)^{49}Ar	2017	(p,pn)	not reported	needs verification	Linh PRC 109 (2024)[22]

Table 4: PDC-side σ_p/p as reported (or not) in public SAMURAI papers. Cells marked "needs verification" or "inferred" should not be filled with invented numbers. The most complete public PDC-side performance numbers are in the Kahlbow 2021 PhD (TUprints 9192)[3]. Storck-Dutine 2023 PhD (TUprints 23768)[17] covers a SAMURAI W/Pt-target campaign with PDC analysis (rather than MINOS LH_2).

The honest reading of Table 4 is that almost every public SAMURAI paper reports physics observables (cross sections, decay energies) but not σ_p/p directly, so the "needs verification" cells are deliberate placeholders for follow-up commits and must never be filled with invented numbers. The Kondo entry is the lone borderline case: the 0.2 MeV FWHM decay-energy resolution lets one *infer* $\sigma_p/p \sim 1\%$ at the order-of-magnitude level, but that figure is inferred and is labelled as such. The recommended path to resolve the remaining cells is the Kahlbow 2021 PhD thesis[3], which contains the most complete public PDC-side performance breakdown; Storck-Dutine 2023[17] is partially relevant but covers a different (W/Pt) target.

4 Gap analysis

4.1 Performance: σ values

The published reference frame sets the bar at the Kobayashi 2013 §3.4 design specification: 1 mm position resolution, 2 mrad angular resolution, and 0.8% relative momentum resolution at the

PDC, with the SAMURAI commissioning RI-beam runs reaching $\Delta p/p \approx 1/1500$[1]. The most recent public physics-grade comparison point is the SEASTAR-III ^{28}O measurement of Kondo et al.[23], which quotes a 0.2 MeV FWHM decay-energy resolution; propagating that to a PDC-side relative momentum resolution gives $\sigma_p/p \sim 1\%$ at the order-of-magnitude level, but the figure is inferred (not directly reported in the paper) and is flagged as such here.

In our own Geant4 + reconstruction chain, the directly observable spatial spread is $\sigma_y(\mathrm{PDC2}) =$ 12.83–14.93 mm in the all-air configuration, 6.80–8.21 mm when Region A (dipole cavity + beam pipe) is set to vacuum (`beamline_vacuum`), and approximately 1.0 mm in the all-vacuum configuration; the corresponding σ_{θ_y} falls in 5.4–9.2 mrad for the all-air and `beamline_vacuum` cases and 2.0–2.6 mrad for the all-vacuum case[6]. We do not introduce new numbers in this subsection; all entries are reproduced from Table 1 in §2.

The position σ_y at PDC2 is therefore an order of magnitude away from the 1 mm hardware design figure in both the all-air and the `beamline_vacuum` configurations, and the angular σ_{θ_y} is 2–4× the 2 mrad design value in those same two configurations. Only the all-vacuum ablation reaches the design floor: $\sigma_y(\mathrm{PDC2}) \approx 1.0$ mm matches the hardware specification to within the ablation precision, and $\sigma_{\theta_y} = 2.0$–2.6 mrad straddles the 2 mrad target. We do not yet have an end-to-end σ_p/p figure from this ablation; the per-method $\sigma_{p,\mathrm{target}}$ propagation methodology under MC, Profile, Laplace, MCMC, and MC-PDC is documented in the RK error-analysis deep-dive memo[9] and is not re-derived here.

The interpretation of this gap structure matters for how follow-up work is scoped. The fact that the all-vacuum ablation lands on the 1 mm hardware design floor is direct evidence that the reconstruction methodology itself is sound: the same RK back-tracking + drift-chamber pipeline that produces the 13 mm number in air produces the 1 mm number in vacuum, with no algorithmic change between the two. The gap to design is therefore primarily a material-budget problem rather than a methodology problem, and the responsibility for closing it sits with §4.3 (material budget and vacuum configuration). The $\sigma_y(\mathrm{PDC2})$ entries propagate backwards through the RK tracker and the SAMURAI field map to a target-momentum uncertainty $\sigma_{p,\mathrm{target}}$, and that propagation is captured method-by-method in the RK deep-dive[9].

4.2 Methodology

The published methodology spine for the SAMURAI proton arm has three pieces. RK back-tracking through the Sato 2013 OPERA-3d / TOSCA field map[10] forms the propagator; the front-end drift-chamber pipeline (drift-time-to-distance, per-plane line fit) is the RIKEN ROOT-based analysis of Otsu et al.[13]; an optional Kalman smoother on the PDC tracks is the standard refinement step but has *no peer-reviewed paper covering the PDC-side methodology*. The closest published Kalman paper is Santamaria et al. Nucl. Instr. Meth. A **905** (2018)[14], which applies a Kalman filter to the MINOS TPC rather than the PDCs. The actual PDC-side Kalman methodology lives only in PhD theses: Santamaria 2015[15], Kahlbow 2021[3], Lehr 2023[16], and Storck-Dutine 2023[17].

Our internal pipeline lives in `libs/analysis_pdc_reco/` (with the repo-level guide in `smsimulator5.5/CLAU` and matches the literature spine piece-for-piece: it provides the same RK back-tracking propagator, exposes multiple selectable RK fit modes (one concrete example being `--rk-fit-mode fixed-target-pdc-only` as exercised in the MS ablation memo[6]), wraps a neural-network backend under `scripts/reconstruction/nn_target_momentum/`, and runs a geometry filter (`libs/qmd_geo_filte` before the fit. The full numerical status of the RK pipeline, including MC, Profile, Laplace, MCMC, and MC-PDC variants, is recorded in the RK status memo[24] and the deep-dive[9].

The methodological gap to the published reference frame is therefore small in algorithm shape: literature and our pipeline use the same RK back-tracking and the same optional Kalman smooth-

[23] Y. Kondo et al., Nature **620**, 965–970 (2023), DOI: 10.1038/s41586-023-06352-6.

[24] See `docs/reports/reconstruction/momentum_reco/rk/rk_reconstruction_status_20260416_zh.pdf` for the per-method RK reconstruction status.

ing. The *actionable* gap is process-noise calibration. None of the four PhD theses or the Santamaria 2018 NIM A paper[14] publishes a process-noise covariance tuned to a specific magnet + air configuration, and there is no public table that maps a given dipole + cavity + beam-pipe + downstream-air configuration to a Kalman process-noise covariance one can drop into our smoother. The quadrature decomposition in the MS ablation memo[6] §6 (with $\sigma_A \approx 11.19$ mm from Region A and $\sigma_B \approx 7.43$ mm from Region B) is exactly the per-region calibration data needed to set process-noise inputs for our specific geometry, and it does not exist in the public literature for this configuration.

The interpretation is that the published reference frame supplies the methodology shape (RK + optional Kalman), but it does not supply the configuration-specific tuning numbers, so we have to self-calibrate the process-noise inputs against our own ablation data rather than read them off a literature table. This is a closeable gap rather than a fundamental obstacle: the calibration data already exists internally[6], and feeding it into the Kalman smoother is the next concrete methodology step.

4.3 Material budget and vacuum configuration

The published reference frame for the SAMURAI vacuum system is Shimizu et al. Nucl. Instr. Meth. B **317**, 739–742 (2013)[12], which documents the dipole cavity as evacuated to a few Pa during the original 2012 commissioning. (We deliberately do not cite this paper for exit-window material: the actual SAMURAI exit window is metallic, and we treat that as a domain fact rather than a paper-derived claim here.) Whether the SEASTAR-III / SAMURAI21 (2017) campaign in particular kept this evacuated-cavity configuration is *inferred from Shimizu 2013 plus standard SAMURAI21 operation*; we do not have a direct quote in the surveyed references that explicitly confirms the Region A vacuum status for SAMURAI21 (2017), and the claim is flagged as inferred rather than quoted.

In our own Geant4 setup, the Region A vacuum status is under exploration rather than fixed. Two candidates from the MS ablation memo[6] §6 frame the choice numerically: with Region A set to vacuum the stage D PDC2 spread is $\sigma_y(\mathrm{PDC2}) = 7.49$ mm, and with Region A set to air it rises to $\sigma_y(\mathrm{PDC2}) = 13.50$ mm. Both candidates are documented and neither is asserted to be the actual experimental run condition.

If the literature default is the evacuated cavity (inferred) and our σ_y under that configuration is 7.49 mm, the residual gap to the 1 mm hardware design figure is dominated by Region B (Exit Window → PDCs, ≈ 2 m of air); if instead our actual configuration is an air-filled cavity giving 13.50 mm, the gap is dominated by Region A. The quadrature decomposition in the same memo[6] §5.2 gives $\sigma_A \approx 11.19$ mm from Region A and $\sigma_B \approx 7.43$ mm from Region B, so Region A air vs. vacuum is the single most consequential configuration question for closing the σ gap.

Per the project's discipline on experimental-reality claims, we deliberately do not assert which of the two Region A configurations corresponds to the actual experimental run; documenting both 7.49 mm and 13.50 mm candidates is the honest position. The actionable next step is collaborator confirmation of the SAMURAI21 (2017) Region A status — this is an engineering / operational question rather than a physics one, and it should be settled by reading the run-time vacuum logs or by direct contact with the SAMURAI operations group rather than by simulation alone.

4.4 Experimental configuration and geometry

The published reference frame for the SAMURAI + MINOS terminal places the reaction target outside the dipole magnet: the MINOS programme of Obertelli et al., EPJA **50**, 8 (2014)[25] proposes a thick LH_2 target surrounded by a TPC vertex tracker upstream of the dipole, with a typical SEASTAR beam energy of ∼ 250 MeV/u; the implementation paper of Santamaria

[25] A. Obertelli et al., Eur. Phys. J. A **50**, 8 (2014), DOI: 10.1140/epja/i2014-14008-y.

et al. NIM A **905**, 138–148 (2018)[14] documents the MINOS TPC tracking and confirms the target-outside-magnet geometry.

In our setup, the deuteron target sits at world coordinates $(-12.488, 0.0009, -1069.458)$ mm and is therefore physically inside the SAMURAI dipole cavity rather than upstream of it[8]. The proton kinematics is fixed at $|\vec{p}| = 627$ MeV/c, corresponding to ~ 190 MeV/u kinetic energy[6]; there is no vertex detector around the target.

Region B (~ 2.0 m, Exit Window $\to$ PDCs) is *directly comparable* to the literature in dipole and PDC1/PDC2 hardware. Any performance number on Region B alone should benchmark against the Kahlbow 2021 PhD[3]. Region A (~ 4.3 m, target-inside-cavity flight) has *no literature precedent*: published SAMURAI+MINOS configurations all place the target outside the magnet, so the in-cavity 4.3 m flight path is a configuration unique to our experiment.

The interpretation is that the geometry difference between literature and our experiment is by design — our experiment puts the target inside the dipole cavity for physics reasons, while MINOS puts it outside for vertex-tracking reasons — and is therefore not actionable in the sense that we should change the geometry to chase a literature number. What it does explain is why no published σ_p/p figure applies to our setup directly: every public benchmark is for a target-outside-magnet configuration, and ours is the first target-inside-magnet case for this PDC chain. The Region B comparison against Kahlbow 2021[3] remains the cleanest benchmark we can run, and Region A has to be self-calibrated against our own ablation data.

Axis	Literature reference	Our state	Action item
Performance σ_y at PDC2	1 mm design (Kobayashi 2013)	$\sim$ 13 mm all-air / $\sim$ 7.5 mm A=vac (ms_ablation §4)	Region A vacuum decision
Methodology	RK + optional Kalman, in PhD theses (Santamaria 2015, Kahlbow 2021)	`libs/analysis_pdc_reco/`	Compare process-noise model to Kahlbow 2021 PhD
Material budget / vacuum	Cavity evacuated, standard SAMURAI (Shimizu 2013)	Region A status under exploration (ms_ablation §6)	Confirm SAMURAI21 (2017) Region A config from collaborators
Geometry	Target outside magnet, MINOS+TPC (Obertelli 2014)	Target inside cavity, no vertex (ms_ablation §1.1)	None — by-design difference

Table 5: Four-axis gap matrix between SAMURAI PDC reconstruction literature and our current state.

5 Broader MINOS+SEASTAR chain context

This section provides bibliographic context only; it does not change the gap analysis in §4.

5.1 SEASTAR programme timeline

The SEASTAR ("Shell Evolution And Search for Two-plus energies At the RIBF") programme had two distinct hardware phases. The first phase, SEASTAR-I and SEASTAR-II, ran on the F8 / ZeroDegree spectrometer and produced the discovery papers on ^{78}Ni-region structure: Santamaria et al., Phys. Rev. Lett. **115**, 192501 (2015)[26], Doornenbal et al., Phys. Rev. C **95**, 041301 (2017)[27], and Taniuchi et al., Nature **569**, 53 (2019)[28].

[26] C. Santamaria et al., Phys. Rev. Lett. **115**, 192501 (2015), DOI: 10.1103/PhysRevLett.115.192501.
[27] P. Doornenbal et al., Phys. Rev. C **95**, 041301 (2017), DOI: 10.1103/PhysRevC.95.041301.
[28] R. Taniuchi et al., Nature **569**, 53 (2019), DOI: 10.1038/s41586-019-1155-x.

The second phase, from approximately 2017 onward, moved the MINOS target into the SAMURAI cave and used the proton arm with the PDC1/PDC2 chain. Representative publications include Cortés et al., Phys. Lett. B **800**, 135071 (2020)[29], Kondo et al., Nature **620** (2023)[30], and Kahlbow et al., Phys. Rev. Lett. **133**, 082501 (2024)[31]. The F8/ZeroDegree campaigns ran before SAMURAI received its proton arm; only the F13/SAMURAI campaigns from ∼2017 onward use the PDC1/PDC2 chain.

5.2 Companion detectors

The PDC1/PDC2 chain co-exists with several other detectors in the SAMURAI cave. Each item below is a one-line role + a citation; this is not an exhaustive sub-system review.

- **MINOS** — liquid hydrogen vertex tracker with TPC, target sits outside the magnet; co-exists with PDC1/PDC2 in the SAMURAI proton arm. A. Obertelli et al., Eur. Phys. J. A **50**, 8 (2014)[25].
- **NeuLAND** — large-area neutron time-of-flight array; co-exists with PDC1/PDC2 in the SAMURAI proton arm. K. Boretzky et al., Nucl. Instr. Meth. A **1014**, 165701 (2021)[32].
- **NEBULA** — plastic-scintillator neutron detector at SAMURAI; co-exists with PDC1/PDC2 in the SAMURAI proton arm. The cross-cutting reference is the SAMURAI/NEBULA recent-developments paper by Kondo, Tomai, Nakamura, Nucl. Instr. Meth. B **463**, 173–178 (2020)[33].
- **DALI2+** — NaI(Tl) γ-ray array; co-exists with PDC1/PDC2 in the SAMURAI proton arm. S. Takeuchi et al., Nucl. Instr. Meth. A **763**, 596 (2014)[34].
- **HODP** — hodoscope plastic for forward charged-fragment identification; co-exists with PDC1/PDC2 in the SAMURAI proton arm. No dedicated NIM publication identified — described in Kobayashi 2013 §3.4[1].
- **CATANA** — Si+CsI charged-particle array; co-exists with PDC1/PDC2 in the SAMURAI proton arm. CATANA NIM publication not located in this survey — described in Kondo et al., Nature **620** (2023) supplementary material[30].
- **STRASSE** — next-generation CH_2 silicon-strip target tracker, in development; co-exists with PDC1/PDC2 in the SAMURAI proton arm. H. N. Liu et al., Eur. Phys. J. A **59**, 125 (2023)[35].

5.3 Reviews and resources

Two review articles cover the broader methodology. T. Aumann, Prog. Part. Nucl. Phys. **118**, 103847 (2021)[36] reviews nuclear reactions with radioactive beams and covers SAMURAI campaigns. V. Panin, T. Aumann, C. A. Bertulani, Eur. Phys. J. A **57**, 103 (2021)[37] reviews quasi-free scattering in inverse kinematics, the methodological frame for MINOS+SAMURAI.

[29] M. L. Cortés et al., Phys. Lett. B **800**, 135071 (2020), DOI: 10.1016/j.physletb.2019.135071.
[30] Y. Kondo et al., Nature **620**, 965–970 (2023), DOI: 10.1038/s41586-023-06352-6. (Same as Table 4.)
[31] J. Kahlbow et al., Phys. Rev. Lett. **133**, 082501 (2024), DOI: 10.1103/PhysRevLett.133.082501.
[32] K. Boretzky et al., Nucl. Instr. Meth. A **1014**, 165701 (2021), DOI: 10.1016/j.nima.2021.165701.
[33] Y. Kondo, T. Tomai, T. Nakamura, "Recent progress and developments for experimental studies with the SAMURAI spectrometer," Nucl. Instr. Meth. B **463**, 173–178 (2020), DOI: 10.1016/j.nimb.2019.05.068.
[34] S. Takeuchi et al., Nucl. Instr. Meth. A **763**, 596 (2014), DOI: 10.1016/j.nima.2014.06.087.
[35] H. N. Liu et al., Eur. Phys. J. A **59**, 125 (2023), DOI: 10.1140/epja/s10050-023-01023-6.
[36] T. Aumann, Prog. Part. Nucl. Phys. **118**, 103847 (2021), DOI: 10.1016/j.ppnp.2021.103847.
[37] V. Panin, T. Aumann, C. A. Bertulani, "Quasi-free scattering in inverse kinematics as a tool to unveil the structure of nuclei," Eur. Phys. J. A **57**, 103 (2021), DOI: 10.1140/epja/s10050-021-00416-9.

A 2024 series of arXiv preprints submitted to PTEP covers direct reactions with hydrogen targets at the RIBF, including: K. Yoshida and J. Tanaka, "Reaction mechanism of quasi-free knockout processes in exotic RI beam era," arXiv:2412.16649[38]; J. Kahlbow, "The southern shore of the island of inversion studied via quasi-free scattering," arXiv:2412.16799[39]; R. Taniuchi, "Competition of the shell closure and deformations across the doubly magic ^{78}Ni," arXiv:2412.16972[40]; and J. Tanaka, M. L. Cortés, H. Liu, R. Taniuchi, "Detectors for next-generation quasi-free scattering experiments at the RIBF," arXiv:2412.17887[41]. The RIKEN Accelerator Progress Report archive[42] contains annual reports including PDC commissioning notes from 2012 onward.

5.4 Caveats from the broader bibliography

"PDC" is RIKEN-internal terminology; most published papers refer to the device as "drift chambers downstream of the SAMURAI magnet" rather than PDC1/PDC2, which complicates literature search. Several papers that look like SAMURAI PDC results on first reading must be excluded from the gap analysis: Kubota et al., Phys. Rev. Lett. **125**, 252501 (2020)[43] uses the dedicated RPD (recoil proton detector), not PDC; Tanaka et al., Science **371**, 260 (2021)[44] was performed at RCNP, not SAMURAI; and SEASTAR-I/II at F8/ZeroDegree predates the SAMURAI proton arm and does not use PDC1/PDC2.

6 Implications and next steps

6.1 Short-term data acquisition action items

The following items are candidate next steps for closing literature-side data gaps; they are scoped to bibliographic retrieval and collaborator queries rather than to new simulations.

- **Fetch the Kahlbow 2021 PhD thesis** from TUprints 9192 (`https://tuprints.ulb.tu-darmstadt.de/9192/`)[3] and extract its σ_p/p, drift-time-to-distance calibration details, and Kalman process-noise specification. The goal is to fill the "needs verification" cells in Table 4 (§3.3) with sourced numbers rather than placeholders.
- **Browse the RIKEN Accelerator Progress Report (APR) archive** for volumes 46–58 (`https://nishina.riken.jp/researcher/APR/index_e.html`)[42] and search for entries by V. Panin, M. Kurokawa, and H. Otsu mentioning PDC commissioning notes. Candidate next step — short bibliographic notes here can plausibly resolve several "needs verification" cells without recourse to a full PhD-length document.
- **Confirm the SAMURAI21 (2017) Region A vacuum configuration** from collaborators or internal run-time records. If confirmed, this closes the open question identified in §4.3 paragraph 4 about whether the literature-default cavity-evacuated configuration matches the SAMURAI21 run; until then, the air-vs-vacuum status remains an open question and both candidates (σ_y(PDC2) = 7.49 mm and 13.50 mm) should continue to be documented side by side.

[38] K. Yoshida and J. Tanaka, arXiv:2412.16649 (submitted to PTEP).
[39] J. Kahlbow, arXiv:2412.16799 (submitted to PTEP).
[40] R. Taniuchi, arXiv:2412.16972 (submitted to PTEP).
[41] J. Tanaka, M. L. Cortés, H. Liu, R. Taniuchi, arXiv:2412.17887 (submitted to PTEP).
[42] RIKEN Nishina Center, *RIKEN Accelerator Progress Report* archive, `https://nishina.riken.jp/researcher/APR/index_e.html`.
[43] Y. Kubota et al., Phys. Rev. Lett. **125**, 252501 (2020), DOI: 10.1103/PhysRevLett.125.252501.
[44] J. Tanaka et al., Science **371**, 260 (2021), DOI: 10.1126/science.abe4688.

6.2 Medium-term simulation and analysis directions

The following items are conditional on the outcome of §6.1 and on the broader RK status; they are explicitly framed as "if confirmed" rather than as commitments.

- **Region A vacuum engineering feasibility.** If a future run configures Region A as vacuum, an engineering-feasibility assessment (vacuum interlocks, cavity pumping, beam-pipe connection to the dipole cavity) becomes the corresponding action item from the gap matrix in Table 5. This step is contingent on the §6.1 collaborator confirmation and is not pursued speculatively.

- **Run `ms_ablation` stage D** feeding $\sigma_A \approx 11.19$ mm and $\sigma_B \approx 7.43$ mm (per[6] §6) as RK process-noise inputs. The goal is to complete the methodology-gap closure identified in §4.2 paragraph 3, where the actionable gap is configuration-specific process-noise calibration rather than algorithm shape.

- **Compare the resulting $\sigma_{p,\mathbf{target}}$ against the literature $\sim 1\%$ figure** once stage D propagates σ_A and σ_B back through the RK tracker and the SAMURAI field map. The reference point is the Kondo 2023 SEASTAR-III decay-energy resolution, from which $\sigma_p/p \sim 1\%$ is inferred at the order-of-magnitude level[30]. The current $\sigma_{p,\text{target}}$ propagation methodology is documented in the RK error-analysis deep-dive[9], with the broader RK status in[24].

6.3 Open questions

The following items are deliberately phrased as questions; none of them is closed by the current document.

- **Is the actual SAMURAI21 (2017) Region A air or vacuum?** This is under exploration. Both candidates are documented numerically: $\sigma_y(\text{PDC2}) = 7.49$ mm if Region A is vacuum, and $\sigma_y(\text{PDC2}) = 13.50$ mm if both Region A and Region B are air (per[6] §6). See §4.3 for the configuration discussion.

- **Is our PDC alignment consistent with the literature standard?** No PDC alignment data or alignment-comparison study has been produced internally to date; this is a follow-up question rather than a settled point, and a literature-vs-our-state alignment comparison has not been attempted in this document.

- **What is the source of our drift-time-to-distance calibration — simulated or measured?** The calibration is currently consumed via the `libs/analysis_pdc_reco/` defaults; the provenance (simulated vs. measured) should be verified before any numerical drift-distance accuracy figure is quoted, and the result compared against the Kahlbow 2021 PhD methodology[3].

6.4 Non-goals

This document deliberately excludes several scopes to keep its boundary tight. It does *not* re-run any `ms_ablation` simulations; it does *not* modify the RK reconstruction code; it does *not* commit to a Region A vacuum decision (both 7.49 mm and 13.50 mm candidates are documented honestly side by side); and it does *not* derive new $\sigma_{p,\text{target}}$ propagation values — those live in the RK error-analysis deep-dive memo[9]. These are deliberate non-goals, not oversights.

7 References

This section is the bibliography for this document. Every entry below carries either a DOI, an arXiv identifier, a thesis-repository handle (TUprints / HAL), or a full URL. Entries whose DOIs were not independently verified at the time of writing are marked [DOI not verified]. Flag tags such as [F13/PDC], [F8-ZDS], [exclude], [F13/PDC, confirmed], [F13/PDC, implicit], [F13/PDC, expected], [partial] indicate the SAMURAI focal-plane configuration and the strength of the PDC linkage for each entry; they follow the convention used throughout this document.

7.1 Hardware and methodology (peer-reviewed, foundational)

- Kobayashi T., et al., "SAMURAI spectrometer for RI beam experiments," Nucl. Instrum. Methods B **317**, 294–304 (2013), DOI: 10.1016/j.nimb.2013.05.089. [F13/PDC]
- Sato H., et al., "Superconducting Dipole Magnet for SAMURAI Spectrometer," IEEE Trans. Appl. Supercond. **23**, 4500308 (2013), DOI: 10.1109/TASC.2012.2237225.
- Shimizu Y., Otsu H., Kobayashi T., Kubo T., Motobayashi T., Sato H., Yoneda K., "Vacuum system for the SAMURAI spectrometer," Nucl. Instrum. Methods B **317**, 739–742 (2013), DOI: 10.1016/j.nimb.2013.08.051. [cite for cavity-evacuated status only; not for exit-window material]
- Otsu H., et al., "SAMURAI in its operation phase for RIBF users," Nucl. Instrum. Methods B **376**, 175 (2016), DOI: 10.1016/j.nimb.2016.02.056.
- Nakamura T. and Kondo Y., review, Nucl. Instrum. Methods B **376**, 156 (2016), arXiv:1512.08380.
- Arai Y., et al., Nucl. Instrum. Methods A **453**, 365 (2000), DOI: 10.1016/S0168-9002(00)00658-6.

7.2 Algorithm references

- Santamaria C., et al., "Tracking with the MINOS TPC," Nucl. Instrum. Methods A **905**, 138–148 (2018), DOI: 10.1016/j.nima.2018.07.053.
- Matta A., et al., "NPTool," J. Phys. G **43**, 045113 (2016), DOI: 10.1088/0954-3899/43/4/045113.
- Panin V., et al., Phys. Lett. B **753**, 204–210 (2016), DOI: 10.1016/j.physletb.2015.11.082. [R3B/LAND, methodological precursor]

7.3 PhD theses with PDC methodology

- Kahlbow J. (2021), TU Darmstadt, TUprints 9192, https://tuprints.ulb.tu-darmstadt.de/9192/. [F13/PDC, confirmed]
- Storck-Dutine S. (2023), TU Darmstadt, TUprints 23768, https://tuprints.ulb.tu-darmstadt.de/23768/. [partial: SAMURAI W/Pt target campaign; PDC content auxiliary]
- Excluded:
  - Santamaria C. (2015), HAL tel-01231191, https://hal.science/tel-01231191. [excluded --- MINOS-focused, F8/ZeroDegree-era]
  - Panin V. (2012), TUprints 3170. [excluded --- R3B/LAND]
  - Atar L. (2015), GSI-2016-00315. [excluded --- R3B/LAND]

7.4 Physics papers using PDC1/PDC2 (confirmed/implicit)

- Kondo Y., et al., Nature **620**, 965–970 (2023), DOI: 10.1038/s41586-023-06352-6. `[F13/PDC, implicit]`
- Kahlbow J., et al., Phys. Rev. Lett. **133**, 082501 (2024), DOI: 10.1103/PhysRevLett.133.082501. `[F13/PDC]`
- Revel A., et al., Phys. Rev. Lett. **124**, 152502 (2020), DOI: 10.1103/PhysRevLett.124.152502. `[F13/PDC, mixed]`
- Holl M., et al., Phys. Rev. C **105**, 034301 (2022), DOI: 10.1103/PhysRevC.105.034301. `[F13/PDC, implicit]`
- Wang H., et al., Phys. Lett. B **843**, 138038 (2023), DOI: 10.1016/j.physletb.2023.138038. `[F13/PDC, implicit]`
- Duer M., et al., Nature **606**, 678 (2022), DOI: 10.1038/s41586-022-04827-6. [F13/PDC, used for α (note exception)]
- Corsi A., et al., Phys. Lett. B **797**, 134843 (2019). `[F13/PDC, implicit]`
- Enciu M., et al., Phys. Rev. Lett. **129**, 262501 (2022), DOI: 10.1103/PhysRevLett.129.262501. `[F13/PDC, implicit]`
- Linh B. D., et al., Phys. Rev. C **109**, 034312 (2024), DOI: 10.1103/PhysRevC.109.034312. `[F13/PDC, implicit]`
- Pohl T., et al., Phys. Rev. Lett. **130**, 172501 (2023), DOI: 10.1103/PhysRevLett.130.172501. `[F13/PDC, implicit]`

7.5 Excluded / context-only

- Kubota Y., et al., Phys. Rev. Lett. **125**, 252501 (2020), DOI: 10.1103/PhysRevLett.125.252501. `[exclude --- uses dedicated RPD not PDC]`
- Tanaka J., et al., Science **371**, 260 (2021), DOI: 10.1126/science.abe4688. `[exclude --- RCNP, not SAMURAI]`
- Taniuchi R., et al., Nature **569**, 53 (2019), DOI: 10.1038/s41586-019-1155-x. `[F8-ZDS, not PDC]`
- Doornenbal P., et al., Phys. Rev. C **95**, 041301(R) (2017), DOI: 10.1103/PhysRevC.95.041301. `[F8-ZDS]`
- Atar L., et al., Phys. Rev. Lett. **120**, 052501 (2018), DOI: 10.1103/PhysRevLett.120.052501. `[exclude --- R3B/LAND at GSI]`
- Panin V., et al., Phys. Lett. B **753**, 204 (2016). `[exclude --- R3B/LAND]`
- Sekiguchi K., et al., Phys. Rev. C **89**, 064007 (2014), DOI: 10.1103/PhysRevC.89.064007. `[exclude --- RIBF dp polarimetry/SMART chamber, not SAMURAI/PDC]`
- Sekiguchi K., et al., Phys. Rev. C **96**, 064001 (2017), DOI: 10.1103/PhysRevC.96.064001. `[exclude --- RIBF dp polarimetry/SMART chamber, not SAMURAI/PDC]`

7.6 Reviews and topical collections

- Aumann T., et al., Prog. Part. Nucl. Phys. **118**, 103847 (2021), DOI: 10.1016/j.ppnp.2021.103847.
- Panin V., Aumann T., Bertulani C. A., Eur. Phys. J. A **57**, 103 (2021), DOI: 10.1140/epja/s10050-021-00416-9.
- Nowacki F., Obertelli A., Poves A., "The neutron-rich edge of the nuclear landscape: Experiment and theory," Prog. Part. Nucl. Phys. **120**, 103866 (2021), DOI: 10.1016/j.ppnp.2021.103866.
- Aumann T. and Bertulani C. A., Prog. Part. Nucl. Phys. **112**, 103753 (2020).
- PTEP Symposium contributions on direct reactions with hydrogen targets at the RIBF (2024 preprints):
  - Yoshida K. and Tanaka J., "Reaction mechanism of quasi-free knockout processes in exotic RI beam era," arXiv:2412.16649 (submitted to PTEP).
  - Kahlbow J., "The southern shore of the island of inversion studied via quasi-free scattering," arXiv:2412.16799 (submitted to PTEP).
  - Taniuchi R., "Competition of the shell closure and deformations across the doubly magic ^{78}Ni," arXiv:2412.16972 (submitted to PTEP).
  - Tanaka J., Cortés M. L., Liu H., Taniuchi R., "Detectors for next-generation quasi-free scattering experiments at the RIBF," arXiv:2412.17887 (submitted to PTEP).

7.7 Construction proposals and internal notes

- RIBF-TAC05 (2005) SAMURAI construction proposal, `https://ribf.riken.jp/RIBF-TAC05/10_SAMURAI.pdf`.
- 2012 SAMURAI Construction Proposal, `https://ribf.riken.jp/SAMURAI/120425_SAMURAIConstProp.pdf`.
- RIKEN tech note `Detector-PDC.pdf`, `https://www.nishina.riken.jp/ribf/SAMURAI/image/Detector-PDC.pdf`.
- SAMURAI tecinfo page, `https://www.nishina.riken.jp/RIBF/SAMURAI/tecinfo.html`.
- RIKEN Accelerator Progress Report archive, `https://nishina.riken.jp/researcher/APR/index_e.html`.

7.8 Project-internal references

- `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_air_20260421_zh.md` — `[superseded, stage A]`
- `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` — `[current, stage A′ rev2]`
- `docs/reports/reconstruction/momentum_reco/rk/rk_error_analysis_deep_dive_20260426_zh.pd`
- `docs/reports/reconstruction/momentum_reco/rk/rk_reconstruction_status_20260416_zh.pdf`
- `docs/reports/reconstruction/track_reco/pdc/pdc_reconstruction_uncertainty_detailed.pdf`
- `smsimulator5.5/CLAUDE.md` and `dpol/CLAUDE.md` — project guides
- `libs/analysis_pdc_reco/` — primary reconstruction framework